DISCOVERING DIFFERENTIALLY CORRELATED SETS ACROSS MULTIPLE TREATMENTS

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ABSTRACT

Discovering Differentially Correlated Sets Across Multiple Treatments

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When comparing data sampled under several different conditions, we often want to study not only the overall treatment effect but also how treatments influence the relationships among variables. Differential Correlation Mining (DCM) is an existing method for identifying groups of variables whose average pairwise correlation differs significantly between exactly two sampling conditions. We extend this framework to the setting of three or more treatments, where the goal is to identify variable sets whose pairwise correlation structure varies across all conditions simultaneously. Our proposed method, VSAT for three or more treatments, generalizes the DCM search procedure using a multi-treatment hypothesis test to iteratively update the variable set of interest, producing sets whose members are more strongly correlated under some conditions than others. We demonstrate the effectiveness of this extension through simulation studies and a real data application, showing that it recovers known differential correlation structure and provides a single unified summary of community-level association patterns across all treatment groups.

Keywords: Select descriptive keywords and separate terms with a comma and a space.

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Chapter 1

INTRODUCTION

Sustainable farming practices are crucial in minimizing the environmental impact of agricultural production. A key factor is understanding soil health, which is essential for crop productivity and longevity. Soil health is largely influenced by the variety and abundance of soil microorganisms, which perform essential functions such as nutrient cycling, decomposition, and disease suppression.

A study conducted by the Chemistry department at California Polytechnic State University, San Luis Obispo sampled soil from a Central California vineyard, measuring microbial community composition across three years and four stages of the grapevine life cycle: bud break, bloom, veraison, and harvest (Watts, Bodwin, & Gilbert, n.d.). Three experiments tested the effects of herbicide presence, cover cropping, and fertilizer type on microbial abundance. Three replicate soil samples were collected per treatment at each stage, yielding 96 total samples. These samples were analyzed for genetic markers leading to an estimate of abundance for each class of microorganism present.

Typically, soil microbial diversity is measured using two complementary metrics. Alpha diversity captures the richness and evenness of taxa within individual samples, while beta diversity measures differences in community composition between samples. Previous analysis of this dataset used permutation tests to assess alpha diversity across treatments, finding no statistically significant differences — suggesting that overall microbial richness and evenness remained relatively constant regardless of treatment. However, some groupwise differences in beta diversity across conditions were detected, indicating possible variation in community interaction of microorganisms in reaction to experimental conditions. Based on these results, this work aims to identify the drivers of these differences in beta diversity.

Figure 1.1: Experiment Design

In this thesis, we introduce a new method, under the umbrella of algorithms we call VSAT (Variable to Set Affinity Testing), motivated by the structure of the soil experiment data. for identifying sets of entities (microorganisms) that are differently correlated across conditions. This new method is a generalization of DCM that identifies variable sets exhibiting condition-dependent correlation structure across three or more treatments simultaneously. In the resulting sets, all members will have the characteristic of having a stronger correlation under some conditions than others.

1.1 Differential Correlation

Comparing data sampled under different conditions typically amounts to a first-order analysis such as a difference in means in two treatment cases, or an ANOVA test in multi-treatment cases. However, these methods often only capture overall behavior across conditions and fail to measure the behavior of individual variables or a small set of variables. Second-order statistics such as correlation uncover these behaviors, leading to a more comprehensive analysis of the treatment on an individual variable level. In the vineyard context, alpha diversity is a first-order summary: it describes the richness or evenness of a single sample in isolation. Beta diversity is second-order: it quantifies the compositional distance between samples, capturing how microbial community structure shifts across treatments rather than simply whether average abundance changes.

Differential correlation allows for a quantification of treatment effects on variables and specifically how different treatments change, or do not change, the relationship between variables. Differential correlation is well defined for a two treatment case, and is simply computed by subtracting the average correlation across the two conditions. For example, given variables i and j, the differential correlation is defined as the difference in their pairwise correlation between two treatment conditions:

$$\Delta r_{ij} = r_{ij}^{(1)} - r_{ij}^{(2)} \tag{1.1}$$

where $r_{ij}^{(1)}$ and $r_{ij}^{(2)}$ denote the Pearson correlation between items i and j under treatments 1 and 2, respectively. A large Δr_{ij} indicates that i and j are strongly correlated under one treatment but not the other, suggesting that the relationship between items changes depending on treatment. Differential correlation importantly contrasts from co-occurrence in the same samples: two items may be consistently present together while showing no differential correlation, or may show strong shifts in correlation structure even at similar abundance levels. As stated above, beta diversity is fundamentally concerned with how microbial community composition differs between samples and treatments. Differential correlation handles this well, as it can describe how relationships between microorganisms shift across treatments through individual Δr values. Applying Equation 1.1 across all pairs of microorganisms produces a complete summary of how the community of organisms responds to treatment.

To gain a deeper understanding of microorganism relationships beyond identifying individual pairs with large differential correlation, a natural question is whether groups of microorganisms have correlations that are stronger under some treatments than others. Such groups would suggest that treatment has a systematic effect on how microorganisms associate with one another, rather than having only an isolated effect on individual pairs. Differential Correlation Mining (DCM) is an existing method for analyzing differences in correlation structure between two different sampling conditions, first introduced by Bodwin, Zhang, & Nobel (2018). DCM identifies groups of variables whose average pairwise correlation is significantly higher under one condition than another. DCM works by iteratively updating a set of interest, at each step computing the average correlation under both conditions and then testing whether each variable of interest is more highly correlated with the current set under one condition than the other. If this is true, the variable is added to the set and the process is repeated. The result of the procedure is an estimated differentially correlated (DC) clique, representing a stable set of variables that have a condition-dependent correlation structure, where correlation between items in the set is stronger or weaker depending on treatment.

Recall that the vineyard data includes three sub-experiments: herbicide (none vs. herbicide), cover crop watering (high, low, and none), and fertilizer type (none, synthetic, and organic). As described above, DCM is well defined for the two-treatment case (herbicide), as a single difference Δr_{ij} has a natural interpretation as the directional shift in correlation from one condition to the other.

For the cover crop and fertilizer experiments, each with three treatment levels, no single Δr_{ij} summarizes the full correlation structure across conditions. One approach is to perform $\binom{3}{2} = 3$ pairwise comparisons, applying DCM separately to each pair. However, this is limited in that results across comparisons may be inconsistent — a DC clique identified in one pairwise comparison may not appear in another, meaning a group of microorganisms may shift their co-occurrence structure between two specific treatments but not others. Rather than a single coherent summary of community response, the three-treatment case yields a collection of comparison-specific cliques that must be interpreted together, making it unclear which contrasts are most meaningful or how to aggregate them into a unified conclusion. This motivates the need for a generalized DCM approach that can handle multiple treatments simultaneously, producing a single summary of differential correlation behavior across all conditions.

Existing methods for differential correlation analysis are largely designed for the two-condition setting and operate at the level of individual variable pairs rather than coherent sets (see Cai & Zhang (2016) and Siska, Bowler, & Kechris (2016)). Extensions to multiple conditions focus on global matrix-level testing rather than set identification

(Gómez, Zhang, & Liu, 2025), testing for global correlation structure differences across treatments. This motivates the need for a generalized DCM approach that handles multiple treatments simultaneously, producing an interpretable set of variables where correlation between items shift coherently across all conditions.

In this thesis, we will propose a generalization of DCM that applies for three-or-more treatments. The approach works almost identically to the two-treatment situation, with the exception of the computed test statistic accounting for all treatment groups simultaneously instead of only a one-to-one comparison. In chapter 2, we will discuss the algorithm in detail.

While the two-treatment formulation of DCM provides a powerful framework for identifying differentially correlated taxa between a pair of conditions, studies like the vineyard soil diversity example, do not always reduce to a one-to-one comparison. Experiments involving seasonal variation, multiple land-use types, or tiered intervention levels naturally produce three or more treatment groups, and applying pairwise DCM repeatedly across all combinations fails to capture the joint structure of correlation differences across the full experimental design. The generalization proposed here addresses this directly by extending the test statistic to measure divergence across all treatment groups in a single unified test, preserving the iterative set-based search strategy of the original algorithm while scaling to more complex experimental structures. This extension is, to our knowledge, the first formulation of DCM that operates natively on three or more treatments.

Chapter 2

VSAT PROCEDURE

2.1 Algorithm Overview

In this section, we discuss the general search procedure used by VSAT for discovering differentially correlated sets of variables. The procedure is nearly identical to the two-treatment DCM algorithm (Bodwin et al., 2018), extended to handle three treatment groups simultaneously. The core logic is the same: the algorithm maintains a candidate set A and refines it iteratively using hypothesis tests that evaluate each item's differential correlation with the current set. My version of the VSAT algorithm differs from DCM in the test statistic, which is generalized to measure variation in correlation structure across all three conditions jointly, and p-value computation.

As noted in Bodwin et al. (2018), an important advantage of DCM, and thus VSAT, is that it produces sets without requiring the user to pre-specify the number or size of output sets. Rather than partitioning taxa into a fixed number of groups as in k-means clustering, or requiring a cutoff as in hierarchical clustering, the algorithm determines set membership entirely through the multiple testing procedure applied at each iteration. An item is included in the output set if and only if it passes the significance threshold, so both the size and composition of the set are driven by the signal in the data rather than by arbitrary tuning choices.

The algorithm begins with an initial set A, which can either be user-specified or selected using a greedy scoring procedure. Initialization is discussed in detail in Section 2.5. The set is then passed through an iterative update step, at each iteration revising A based on hypothesis tests for differential correlation intensity across all treatment groups. Hypothesis testing is done based on a user-specified method, choosing between two possible testing approaches:

a Kruskal-Wallis approach, discussed in Section 2.3, and a permutation-based approach, discussed in Section 2.4. The motivation for both approaches and the formal hypothesis being tested are discussed in Section 2.2, where we address why standard parametric testing is not directly applicable to correlation-based statistics of this form. Regardless of the chosen method, p-values are adjusted for multiple comparisons using the Benjamini-Hochberg procedure for dependent tests (Benjamini & Yekutieli, 2001), and items are retained whose adjusted p-values fall below a chosen significance level α.

Iteration continues until the set stabilizes, yielding an estimated differentially correlated (DC) set. These DC sets have the desired characteristic discussed in Section 1: a group of items where correlation between items in the set have differing strength across conditions.

Detailed below is the pseudo-code for the VSAT algorithm:

2.2 Problems with a Parametric Approach

As discussed previously, at each iterative step, A is updated using a hypothesis test. When comparing the effect of three different treatments, the common practice is to use a standard one-way ANOVA test, which assesses whether the mean of some quantity differs across groups. ANOVA requires that the quantity being compared is approximately normally distributed within each group. In the context of our problem, the natural quantity to compare across treatment groups is the average correlation between item j and the current set A under each condition. We define this formally as:

$$S_k(j, A) = \frac{1}{|A \setminus j|} \sum_{i \in A \setminus j} r_k(i, j) \tag{2.1}$$

where $r_k(i, j)$ is the sample correlation between items i and j under condition k. We call this quantity the "connection" between every item $i \in A \setminus j$ and j. To measure how much $S_k(j, A)$ varies across the three conditions simultaneously, we define the affinity of j with respect to A as:

$$\Delta(j, A) = \frac{1}{3}\sum_{k=1}^{3}(S_k(j, A) - \bar{S}(j, A))^2 \tag{2.2}$$

where $\bar{S}(j, A) = \frac{1}{3}\sum_{k=1}^{3} S_k(j, A)$ is the average of the group-specific mean correlations. This parallels the between-group sum of squares in ANOVA, where $S_k(j, A)$ plays the role of a group mean and $\bar{S}(j, A)$ plays the role of the overall mean. A large value of $\Delta(j, A)$ indicates that the average correlation between j and the current set differs across conditions, making j a candidate for inclusion in the DC set.

However, applying ANOVA directly to test whether $\Delta(j, A)$ is significantly large requires that $S_k(j, A)$ is approximately normally distributed within each group. Sample correlations are known to be non-normally distributed, particularly in small samples, since their distribution is bounded between -1 and 1 and is skewed when the true correlation departs from zero. The plot below exemplifies this:

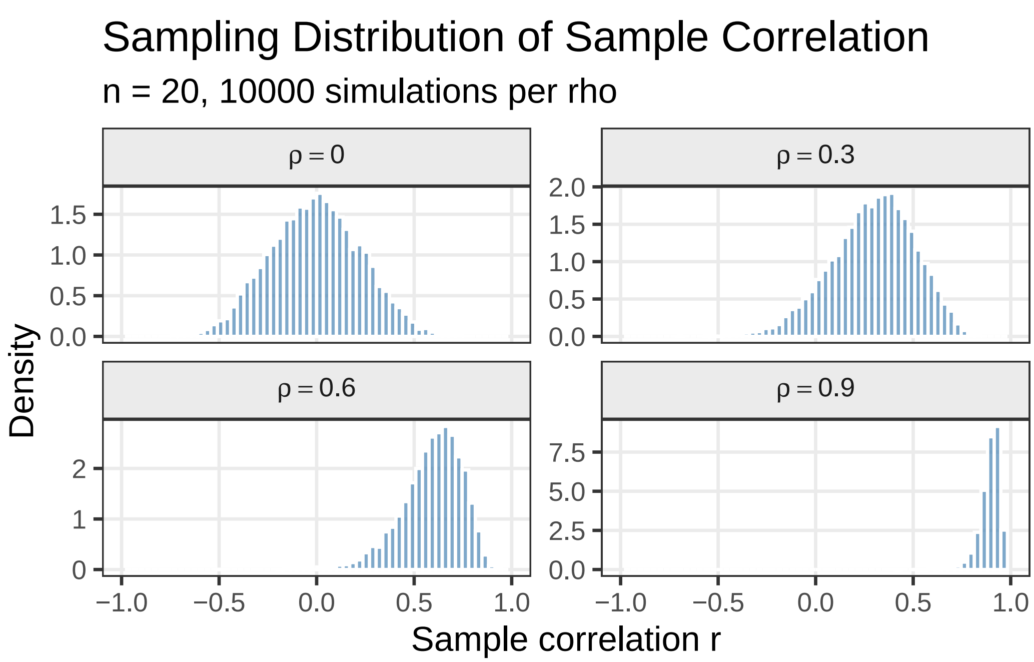

Figure 2.1

Since $S_k(j, A)$ is an average of sample correlations across the members of A, it inherits this same non-normality, and the problem is most severe precisely in the setting we care about, where the true correlation within A is large under at least one condition. Figure 2.2 confirms this using the simulation framework described in detail in Section 3.

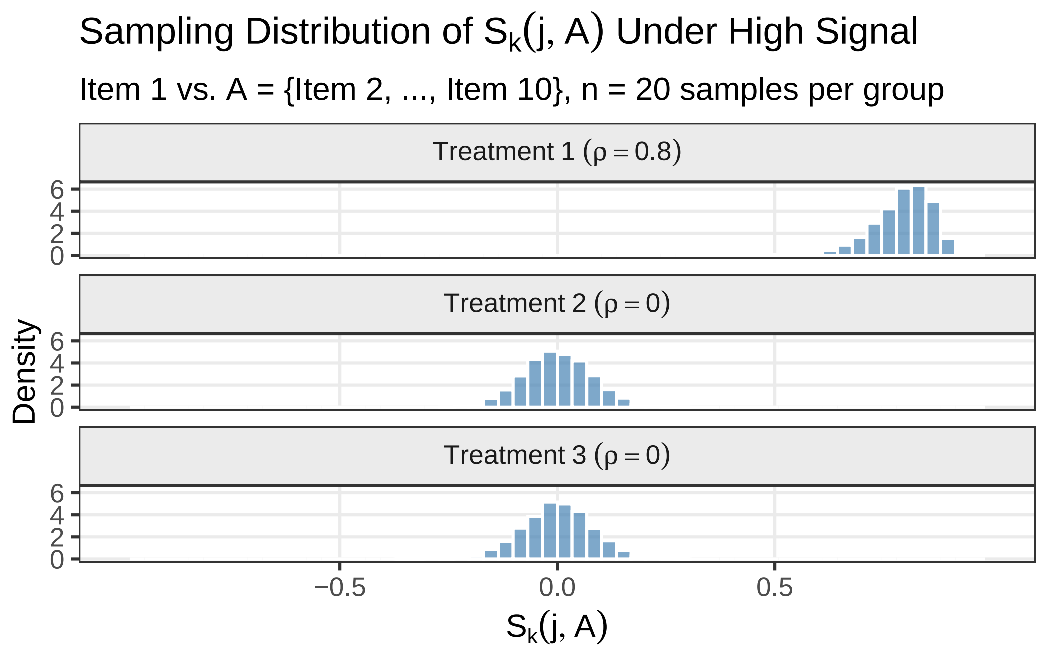

Figure 2.2

Under the high signal scenario, where Treatment 1 has true within-set correlation $\rho = 0.8$ and Treatments 2 and 3 have $\rho = 0$, the distribution of $S_1(j, A)$ is strongly left-skewed while $S_2(j, A)$ and $S_3(j, A)$ remain more symmetric. A normal approximation fits poorly for Treatment 1, where it matters most.

To combat this issue, Bodwin et al. (2018) leveraged individual sample-level correlation contributions to construct a test statistic that had approximately normal behavior in the two-treatment situation, despite sample correlations being highly non-normal. Contributions refer to the inner products of each observation's standardized measurements with the mean profile of the current set A under each condition. They show that their resulting test statistic, when standardized by a consistent estimator of its null variance, follows an approximate standard normal distribution for sufficient sample sizes. Using this fact, items can be tested with a parametric approach for inclusion in a DC set. Based on this previous result, a natural starting point for the three-treatment case is to construct an analogous test statistic using the same sample-level correlation contributions, now computed across all three conditions simultaneously.

We define the necessary quantities for the three-treatment correlation contribution framework. First, we standardize each item's measurements by row within each treatment group. For item i and sample j, define:

$$u_{ij} = \frac{x_{ij} - \bar{x}_i}{\mathrm{sd}(x_i)}, \quad v_{ij} = \frac{y_{ij} - \bar{y}_i}{\mathrm{sd}(y_i)}, \quad w_{ij} = \frac{z_{ij} - \bar{z}_i}{\mathrm{sd}(z_i)}$$

these depict the raw measurements of item i in sample j under treatments 1, 2, and 3 respectively. Standardization is performed across samples within each item and each treatment group, so each $\vec{u}_i$, $\vec{v}_i$, $\vec{w}_i$ is centered at zero. This is done for computational convenience and clarity because we are interested in pairwise correlations rather than raw values. Recall we define the connection between an item and the current set A to be the average correlation between item j and each item $i \in A \setminus j$ under each treatment condition, shown in Equation 2.1 as:

$$S_k(j, A) = \frac{1}{|A \setminus j|} \sum_{i \in A \setminus j} r_k(i, j)$$

To uncover the individual contributions to correlation made by each item to the connection for a given treatment, we can expand Equation 2.1 into standardized values. The connection in treatment 1 can be expanded:

$$S_1(j, A) = \frac{1}{|A \setminus \{j\}|} \sum_{i \in A \setminus \{j\}} r_k(i, j) = \frac{1}{|A \setminus \{j\}|} \sum_{i \in A \setminus \{j\}} \frac{1}{n_1} \sum_{\ell=1}^{n_1} u_{j\ell} u_{i\ell} = \frac{1}{n_1} \sum_{\ell=1}^{n_1} u_{j\ell} \left(\frac{1}{|A \setminus \{j\}|} \sum_{i \in A \setminus \{j\}} \right. \tag{2.3}$$

Define $m_{1\ell}$ is the mean of the ℓ-th sample across all items in $A \setminus \{j\}$ under treatment 1, so that:

$$S_1(j, A) = \frac{1}{n_1} \sum_{\ell=1}^{n_1} u_{j\ell} \left(\frac{1}{|A \setminus \{j\}|} \sum_{i \in A \setminus \{j\}} u_{i\ell} \right) = \frac{1}{n_1} \sum_{\ell=1}^{n_1} u_{j\ell} m_{1\ell}^{(A)} = \mathrm{Cov}(\vec{u}_j, \vec{m}_1^{(A)})$$

Contributions to $S_1(j, A)$ are then the vector $\tilde{u}_{jl} := \{u_{j1}m_{11}^{(A)}, u_{j2}m_{12}^{(A)}, ..., u_{jn_1}m_{1n_1}^{(A)}\}$ with $S_2(j, A)$ and $S_3(j, A)$ being defined using their corresponding contributions as $\tilde{v}_{jl} := \{v_{j1}m_{11}^{(A)}, ..., v_{jn_2}m_{1n_2}^{(A)}\}$, $\tilde{w}_{jl} := \{w_{j1}m_{11}^{(A)}, ..., w_{jn_3}m_{1n_3}^{(A)}\}$.

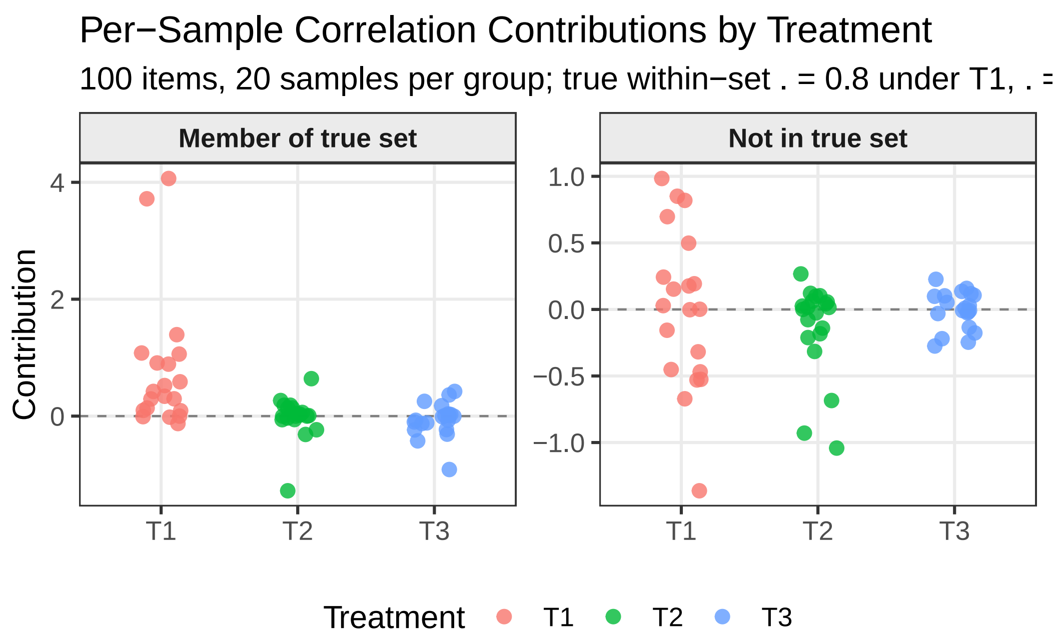

Figure 2.3

As illustrated in Figure 2.3, for an item inside the true set the contributions differ visibly across treatments, whereas for an item not in the set, there is no clear difference in average value across the treatments. This mirrors the structure of a one-way ANOVA: the $\tilde{u}_{j\ell}, \tilde{v}_{j\ell}, \tilde{w}_{j\ell}$ values play the role of within-group observations, $S_k(j, A)$ plays the role of group means, and $\bar{S}(j, A)$ plays the role of the overall mean. The between-group variation present for in-set items but absent for out-of-set items is precisely what an ANOVA-like test is designed to detect, motivating the construction of an F-statistic from the between and within-group variation of the contributions.

Formally, these results can be represented using an ANOVA sum of squares structure:

$$SS_{treatment} = \sum_{k=1}^{3}(S_k(j, A) - \bar{S}(j, A))^2$$

$$SS_{error} = \sum_{\ell=1}^{n_1}(\tilde{u}_{j\ell} - S_1(j,A))^2 + \sum_{\ell=1}^{n_2}(\tilde{v}_{j\ell} - S_2(j,A))^2 + \sum_{\ell=1}^{n_2}(\tilde{w}_{j\ell} - S_3(j,A))^2$$

Under the ANOVA assumptions, which would require contributions are approximately independent, normally distributed, and have equal variance across treatment groups the ratio:

$$F = \frac{SS_{treatment}/(3-1)}{SS_{error}/(n_{total}-3)} \approx F_{2,n_{total}-3}$$

This, in theory, would be the multivariate version of the correlation contribution breakdown from Bodwin et al. (2018), which would allow for the use of a parametric test in the update step. However, the contributions $(\tilde{u}_{j\ell}, \tilde{v}_{j\ell}, \tilde{w}_{j\ell})$ must be independent and normally distributed for the statistic to reliably follow the F-distribution.

We implemented this approach, using the parametric F-test to iteratively update the candidate set A, and applied it to simulated data. Based on our studies, we found the results to be untrustworthy. Computed p-values were all highly insignificant, which in the context of VSAT, which in the context of VSAT always produced empty candidate sets regardless of the true signal level or set size. This prompted an examination of the assumptions underlying the F-statistic.

We know independence does not hold, as dependence is introduced by the row standardization. For example, computing $u_{j\ell} = \frac{(x_{j\ell} - \bar{x}_j)}{\text{sd}(x_j)}$ involves all n_1 samples, so contributions within a group are not independent of one another. We can assess the normality assumption using simulated data from the framework described in Section 3. Using a simulation with 100 items, 20 samples per group, and a true correlation between items in the target set of $\rho = 0.8$ under treatment 1 and $\rho = 0$ under treatments 2 and 3, Figure 2.4 displays the individual contributions $\tilde{u}_{j\ell}, \tilde{v}_{j\ell}, \tilde{w}_{j\ell}$ for an item known to be in the target set:

As shown in Figure 2.4, the normality assumption also clearly fails. Under Treatment 1, where the true within-set correlation is $\rho = 0.8$, the contributions are strongly right-skewed. With both the independence and normality assumptions violated, the F-statistic does not

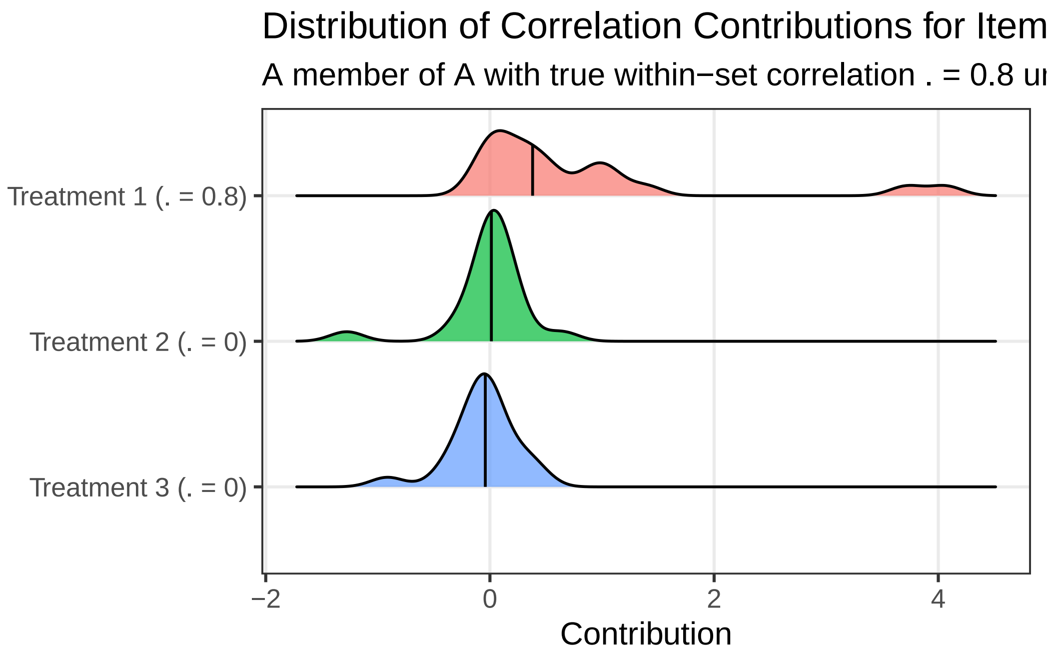

Figure 2.4

follow the $F_{2,n_{total}-3}$ distribution under H_0, which explains the uniformly large p-values observed in practice.

However, the parametric approach was not entirely uninformative. Although p-values were inflated, those corresponding to items that should have been included in A were consistently smaller than those for items outside the true set. This indicated that the correlation contributions were carrying the appropriate information, and that the breakdown was due to violated assumptions rather than a fundamental flaw in the contributions. This motivates the two alternatives discussed in Section 2.3 and Section 2.4 to bypass assumptions with non-parametric methods.

2.3 Kruskal-Wallis Testing

The Kruskal-Wallis test (Kruskal & Wallis, 1952) is a non-parametric analog of one-way ANOVA that assesses whether the distributions of some statistic differ across three or more groups based on the ranks of the observed values. Unlike the F-test, it does not require the observations to be normally distributed or have equal variance across groups, making it a natural substitute in our setting, since correlations are highly non-normal under high signal. The Kruskal-Wallis test does still assume approximate independence of observations across groups, an assumption we noted in Section 2.2 is not strictly satisfied here. This assumption was also not strictly satisfied in the original DCM procedure, so we will proceed under the same justification.

Using the Kruskal-Wallis test, we assess the following null hypothesis for determining set inclusion of an item j into a candidate set A, evaluated at some arbitrary iteration of the VSAT procedure:

$$H_0 : S_1(j, A) = S_2(j, A) = S_3(j, A)$$

We test this against the alternative that average correlation between item j and items in set A differs across at least one pair of treatment conditions. To test these hypotheses, we apply the Kruskal-Wallis test to the correlation contributions that were defined in Section 2.2, pooling all $n_{total} = n_1 + n_2 + n_3$ contributions, ranking them, and computing:

$$H = \frac{12}{n_{total}(n_{total} + 1)} \sum_{k=1}^{3} \frac{R_k^2}{n_k} - 3(n_{total} + 1)$$

where R_k is the sum of ranks assigned to contributions in treatment k. Under H_0, for sufficiently large sample sizes, H approximately follows a χ^2 distribution with 3 - 1 = 2 degrees of freedom, from which a p-value is obtained. The resulting p-value is adjusted using the Benjamini-Hochberg procedure and evaluated for inclusion in A at the chosen significance level.

2.4 Permutation Testing

A second, alternative non-parametric approach is using permutation testing. Permutation testing is well established for use in correlation analysis and has been shown to be effective in determining overall differences in correlation structure across conditions (Gómez et al., 2025). The advantage to using the permutation testing method is that it makes no assumptions about the data. Unlike the Kruskal-Wallis test, it also does not require independence of observations on top of making no distributional assumptions about the data. This makes it particularly well suited to situations involving correlation-based statistics, where independence is difficult to establish.

Our implementation of permutation testing assesses the same hypotheses outlined in Section 2.3, but does not utilize contributions. Instead, we construct the statistic based on the affinity directly, identical to Equation 2.2:

$$\Delta(j, A) = \frac{1}{3}\sum_{k=1}^{3}(S_k(j, A) - \bar{S}(j, A))^2$$

To construct the empirical null distribution sample labels are randomly reassigned across treatment groups while preserving the original group sizes, eliminating any true condition-specific correlation structure while maintaining the overall variability of the data. For each permutation, the affinity $\Delta(j, A)$ is recomputed on the permuted datasets, and the process is repeated many times to produce a null distribution of affinity (□) values under H_0. The p-value for item j is determined based on the proportion of permuted affinities at least as extreme as the observed value.

$$\hat{p}_j = \frac{1}{B}\sum_{b=1}^{B}\left[\Delta^{(b)}(j, A) \geq \Delta(j, A)\right]$$

where B is the specified number of permutations (default B = 1000).

As with the Kruskal-Wallis approach, the resulting p-values are adjusted using the Benjamini-Hochberg procedure before determining inclusion in the candidate set at the chosen significance level.

2.5 Initialization

VSAT relies on a current set A at each iteration, including the first, for item comparison. The choice of initial set therefore determines the starting point of the search procedure. We offer two options for set initialization.

The first option is a user-specified set of interest. This is the preferred option when prior knowledge about the system is available, such as biological hypotheses about which items may exhibit differential correlation, or a set identified from a previous analysis. A user-specified initialization tends to produce more reliable results, particularly when the true differentially correlated set has only weak signal or when only a subset of the true members are included in the initial set. For example, a researcher interested in whether a particular group of microbial taxa exhibits condition-dependent co-occurrence behavior can initialize A with that group directly, allowing the algorithm to refine the set from a starting point motivated by domain knowledge.

The second option is a greedy initialization procedure. All items are ranked by the variance of their mean standardized value across treatment groups as a cheap way of identifying items whose behavior differs across conditions, keeping only the top M items as a candidate starting point. M can be adjusted based on the size of the overall item set. For each item j in the top M, the chosen testing method is applied treating item j as a singleton set. The top items with the most significant resulting p-values are collected as a candidate initial set, and the process is repeated for all M starting points. The candidate initial set with the smallest adjusted p-values are kept as the set passed into the first iteration of the main VSAT loop. We chose to employ a greedy initialization because we found that, in practice, beginning from a random or empty start almost always leads to an empty resulting set.

Chapter 3

SIMULATION STUDY

To evaluate the performance of VSAT, we conduct a simulation study where the true differentially correlated set is known prior to applying the algorithm. We compare the resulting sets formed by VSAT to the true known sets, providing us with a basic understanding of performance under different controlled conditions.

3.1 Simulation Specifications

Data are generated as multivariate normal samples across three treatment groups, each with p items and n samples. A single true DC set is specified prior to generation, with items in the set assigned a common pairwise correlation of ρ_k under treatment k, and all items outside the true set generated as independent standard normal noise with no correlation structure. The differential signal is controlled entirely through the choice of (ρ_1, ρ_2, ρ_3). For example, setting $\rho_1 = 0.8$ and $\rho_2 = \rho_3 = 0$ produces a dataset where items in the set are strongly correlated under treatment 1 but uncorrelated under treatments 2 and 3. Across the study, we varied the number of items p, the number of samples per group n, the size of the true set, and the correlation strengths (ρ_1, ρ_2, ρ_3). Since the true differentially correlated set is known by construction in the simulation, we can evaluate VSAT using precision, recall, and F1 score, comparing the returned set to the known ground truth.

3.2 Simulation Results

We first assess how strongly the differential correlation signal must be for VSAT to reliably recover the true DC set, fixing p = 100, n = 36 and the true set size at 10, while varying ρ_1

between 0.3 and 0.9, keeping $\rho_2 = \rho_3 = 0$ constant. We do not specify a starting set in this case, allowing for greedy initialization of A.

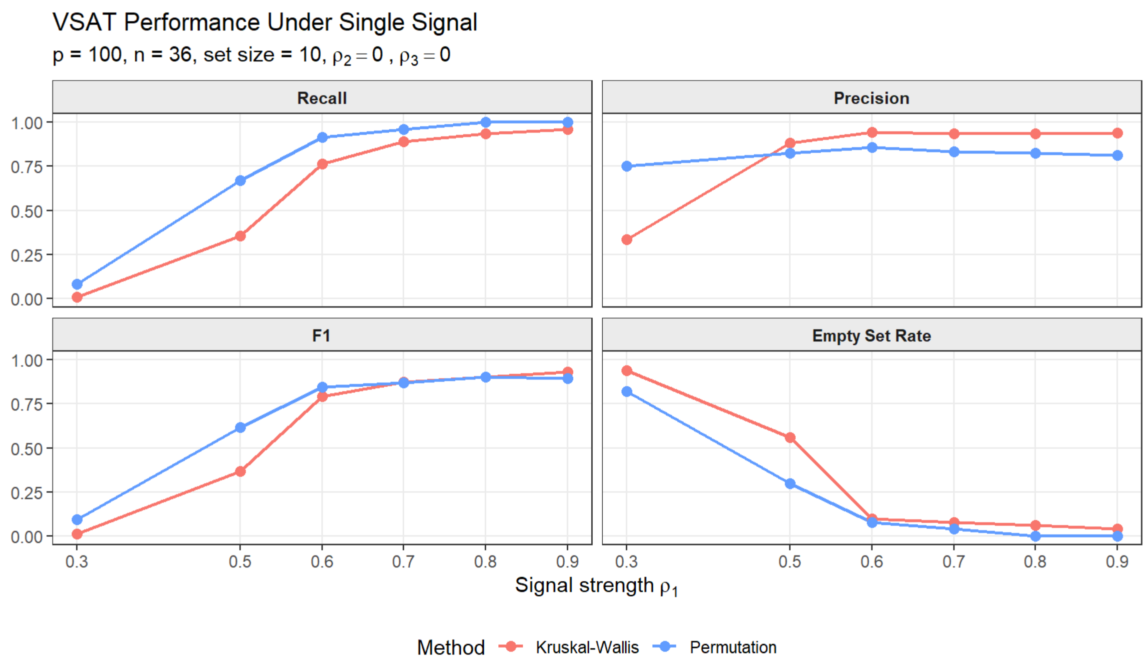

Figure 3.1

Both methods show consistent improvement across all metrics as signal strength increases. At low signal ($\rho_1 = 0.3$), both methods fail to detect the true set in nearly all runs, with empty set rates of approximately 0.97 for Kruskal-Wallis and 0.82 for the permutation method. The permutation method shows a meaningful advantage beginning at $\rho_1 = 0.5$, achieving roughly twice the recall of Kruskal-Wallis and a substantially lower empty set rate. Both methods become largely reliable by $\rho_1 = 0.6$, with recall above 0.75 and empty set rates near zero. At high signal, the methods diverge in precision: Kruskal-Wallis achieves precision near 0.95 at $\rho_1 = 0.9$, while the permutation method stabilizes around 0.80, indicating it tends to include more items outside the true set. Overall, both methods require a within-set correlation of at least $\rho_1 = 0.6$ to reliably detect the true set, with the permutation method preferred when signal is weak and Kruskal-Wallis preferred when signal is strong.

Under the signal gradient scenario, $\rho_2 = 0.3$ and $\rho_3 = 0.1$ are held constant while ρ_1 varies between 0.3 and 0.9, with all other parameters matching the single signal study. When ρ_1 = 0.3, treatments 1 and 2 have identical within-set correlation, so there is effectively no differential signal to detect. Set discovery should increase as ρ_1 grows further from the background signal of $\rho_2 = 0.3$.

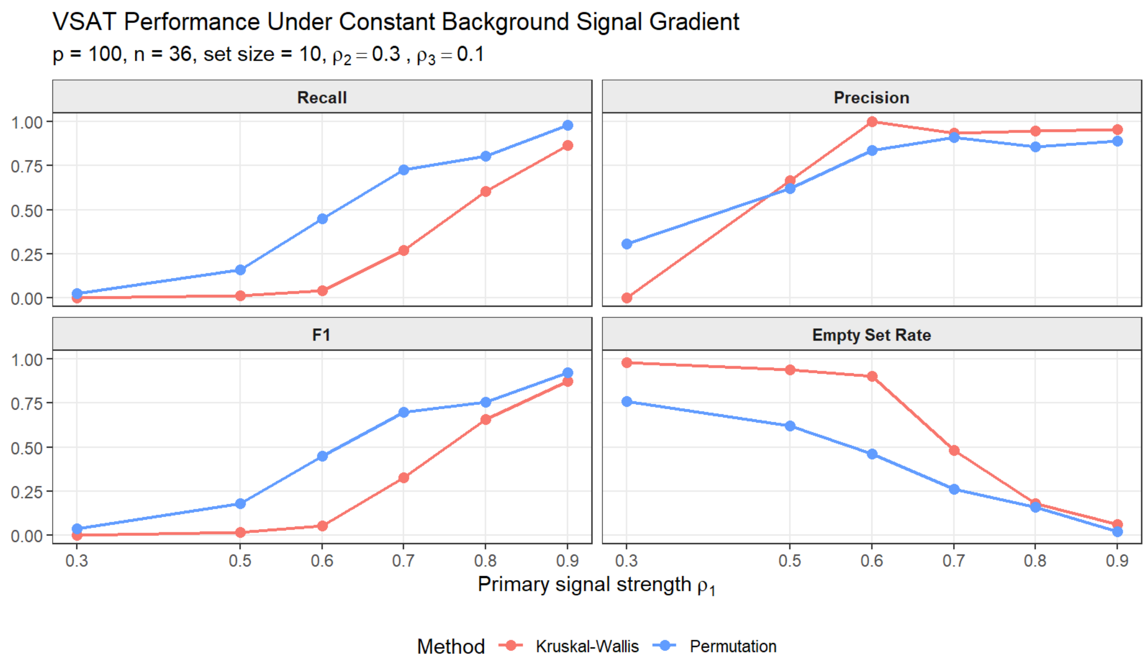

Figure 3.2

As shown in Figure 3.2, the permutation method outperforms Kruskal-Wallis across all signal levels in terms of recall, F1, and empty set rate. The gap between methods is considerably larger here than in the single signal scenario, suggesting that the Kruskal-Wallis test is particularly sensitive to the presence of correlation across all treatments simultaneously, making it less able to distinguish differential from uniform correlation. The permutation method shows meaningful recall beginning around ρ_1 = 0.7, reaching near-perfect recovery by ρ_1 = 0.9, while Kruskal-Wallis does not achieve comparable performance until ρ_1 = 0.9. The one setting where Kruskal-Wallis is competitive is precision at high signal (ρ_1 > 0.7), where it slightly outperforms the permutation method, consistent with the pattern observed in the single signal study.

Together, the single signal and gradient results suggest that the permutation method is the more robust choice across a range of conditions, particularly when the differential signal is moderate or when background correlation is present. Kruskal-Wallis achieves comparable or slightly better precision at high signal but at the cost of substantially reduced sensitivity elsewhere. We additionally assess how VSAT performance varies with the number of samples per group n, fixing $\rho_1 = 0.7$, $p = 100$, and the true set size at 10.

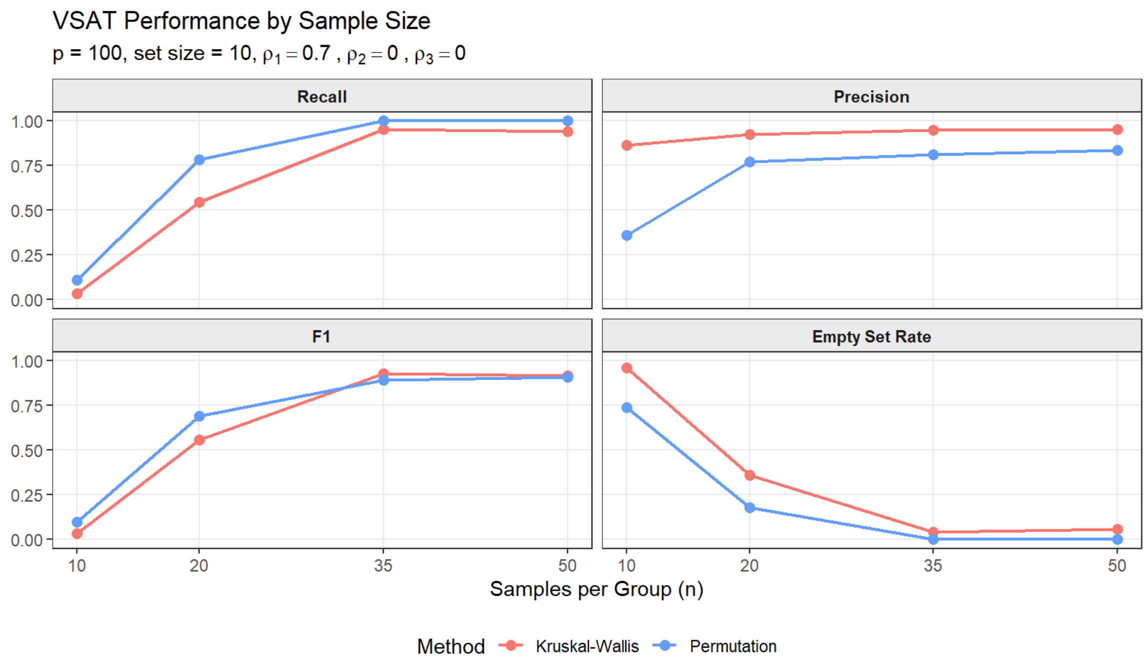

Figure 3.3

As expected, performance improves with larger sample sizes. At $n = 10$, both methods show high empty set rates, approximately 0.97 for Kruskal-Wallis and 0.75 for the permutation method, and near-zero recall, reflecting the difficulty of estimating reliable correlations from very few samples. Performance improves rapidly between $n = 10$ and $n = 20$, with the permutation method achieving recall above 0.75 and an empty set rate near 0.19 at $n = 20$, while Kruskal-Wallis lags behind with recall of approximately 0.55 and an empty set rate of 0.35. By $n = 35$, both methods are largely reliable, with high recall and empty set rates close to zero. The permutation method shows a consistent advantage in recall at small sample sizes, while Kruskal-Wallis achieves slightly higher precision throughout, consistent with the signal strength results.

Finally, we assess performance as a function of the total number of items pp p, fixing $\rho_1 = 0.7$, $n = 36$, and the true set size at 10.

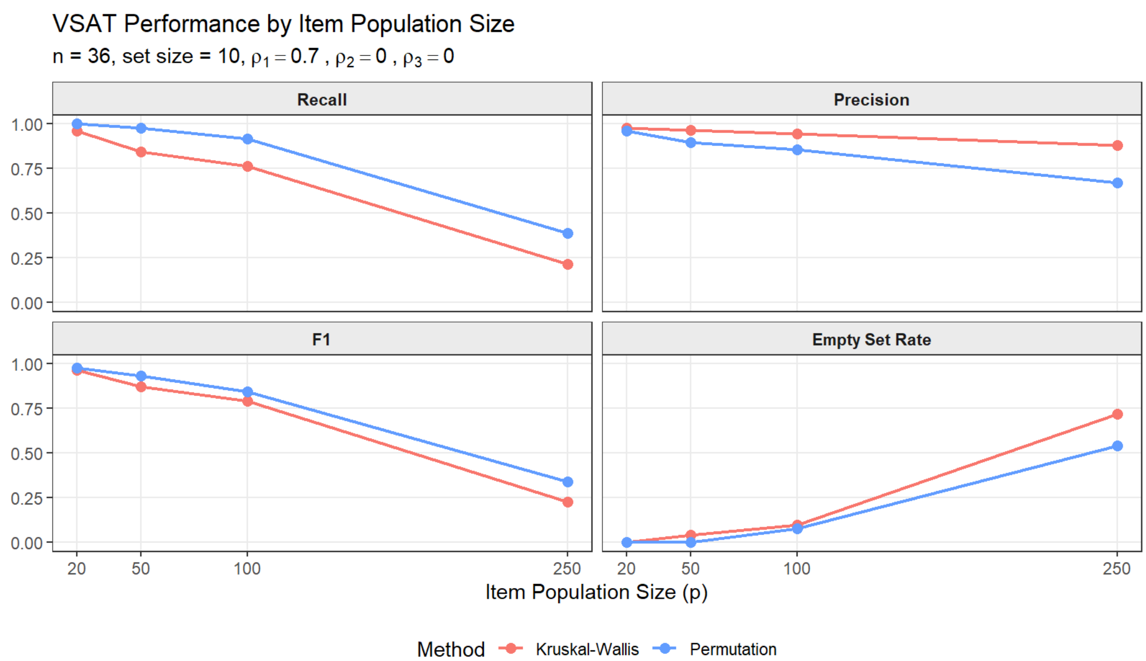

Figure 3.4

The population size results reveal an important scalability limitation. As p grows from 20 to 250, recall and F1 degrade steadily for both methods, with empty set rates rising sharply at $p = 250$, reaching approximately 0.72 for Kruskal-Wallis and 0.55 for the permutation method. This degradation occurs because a larger item population introduces more candidates for false inclusion, making the multiple testing correction increasingly stringent, so items that would have been significant at $p = 100$ may no longer pass the adjusted threshold at $p = 250$. The permutation method degrades more gracefully in recall and empty set rate as p increases, though Kruskal-Wallis maintains higher precision at large p, dropping only to approximately 0.87 at $p = 250$ compared to 0.68 for the permutation method.

Taken together, these results suggest that VSAT is most reliable when sample sizes are at least 35 per group and the item population does not greatly exceed 100 items at moderate signal strengths. Beyond $p = 100$, performance degrades noticeably, with the permutation method being preferred for recall but Kruskal-Wallis maintaining higher precision where computational cost permits.

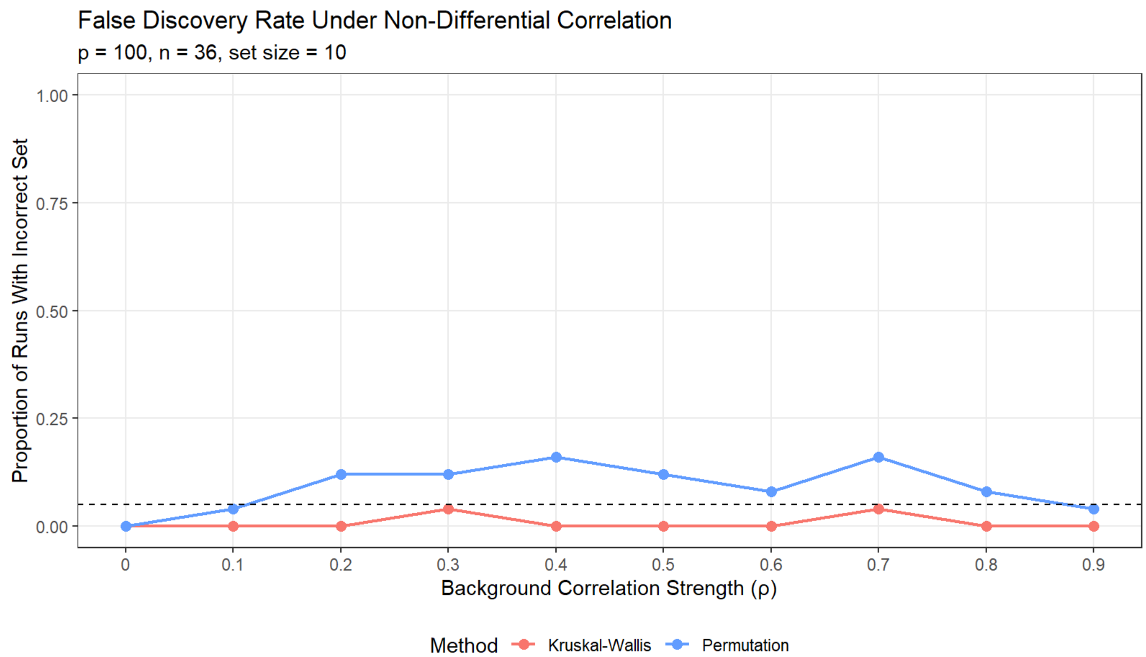

Figure 3.5

Figure 3.5 shows the false discovery rate under non-differential correlation, where $\rho_1 = \rho_2 = \rho_3 = \rho$ and a returned DC set is counted as a false discovery. The Kruskal-Wallis method shows very low false discovery rates across all background correlation levels, remaining at or below the conventional 0.05 threshold. The permutation method shows a more elevated rate, ranging from roughly 5-15% across correlation levels, consistently above the 0.05 threshold. These estimates should be interpreted with caution given only 25 simulation replications per condition due to time restrictions and consistency. as the zero rates observed for Kruskal-Wallis are consistent with a true underlying rate anywhere up to approximately 12% and should not be taken as evidence of perfect null control. Despite the low number of replicates, the difference between methods is meaningful: the Kruskal-Wallis method appears to control false discoveries considerably better than the permutation method under non-differential correlation, consistent with its more conservative trends observed throughout the simulation study.

3.3 Time/Memory Cost

We benchmark VSAT to compare Kruskal-Wallis and permutation methods across the number of items p and samples per group n. Both benchmarks use $n_{perm} = 200$ permutations for the permutation method.

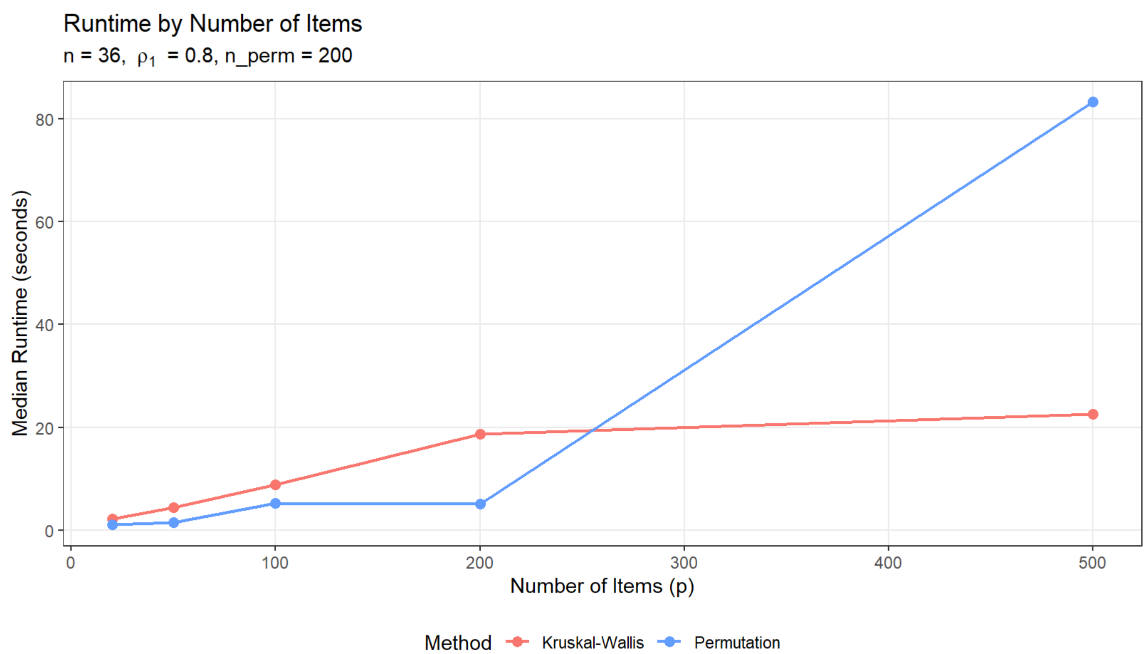

Figure 3.6

Figure 3.6 shows runtime as a function of p. The Kruskal-Wallis method scales approximately linearly, growing from roughly 2 seconds at $p = 20$ to roughly 23 seconds at $p = 500$. The permutation method, by contrast, was faster than the Kruskal-Wallis method for $p \leq 200$, but grows sharply beyond that, reaching approximately 83 seconds at $p = 500$. This severe scaling issue is expected, as each permutation requires computing three $p \times p$ correlation matrices ($O(p^2 n)$), repeated n_{perm} times per VSAT iteration.

Figure 3.7 shows runtime as a function of n, the sample size. Both methods scale with n, as larger samples increase the cost of both row-scaling (Kruskal-Wallis) and correlation matrix

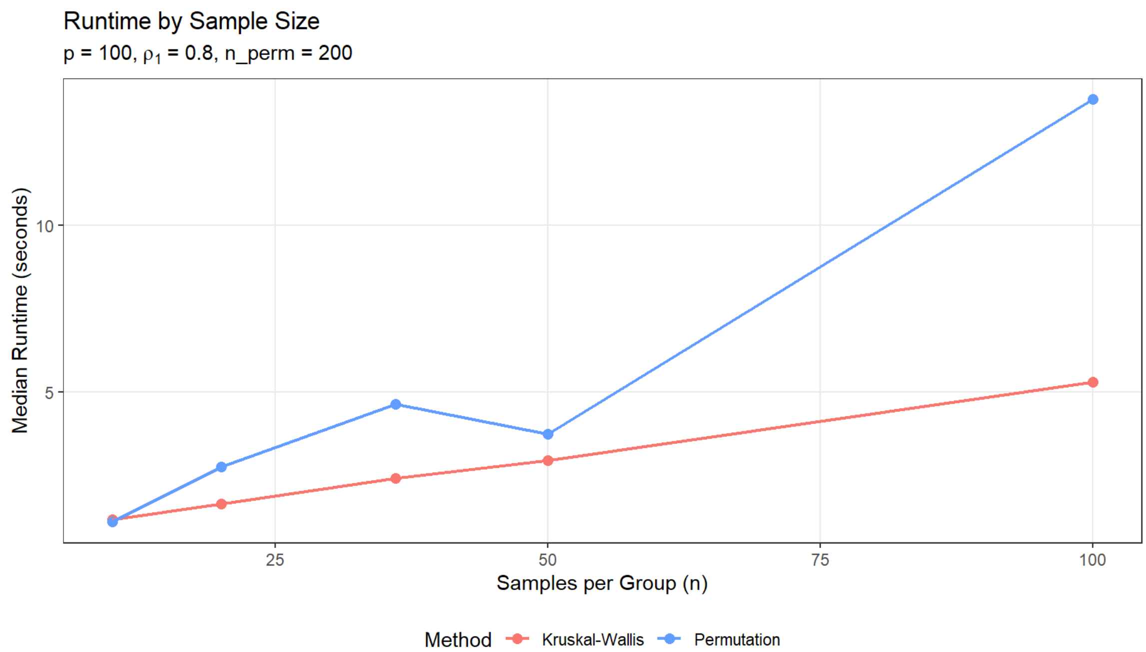

Figure 3.7

computation (permutation). However, the permutation method grows more steeply than the Kruskal-Wallis method with sample size. Therefore, Kruskal-Wallis method is preferred in terms of time cost when p is large, while the permutation method is competitive for smaller datasets where its exact null distribution may be worth the additional cost.

A natural question is how much of this runtime is attributable to initialization versus the core iterative procedure. We examine this next by comparing greedy initialization against runs supplied with a pre-specified anchor set A_0, which bypasses initialization entirely and isolates the cost of the iterative loop alone.

Figure 3.8 compares runtime under greedy initialization against a pre-specified set of three confirmed true DC taxa, across both methods. For the Kruskal-Wallis method, providing a pre-specified set reduces runtime to near zero across all values of p, confirming that initialization dominates its computational cost. The iterative loop itself is essentially free once a starting point is supplied. The permutation method tells a very different story: the informed starting point is substantially slower than greedy, reaching approximately 460 seconds at p

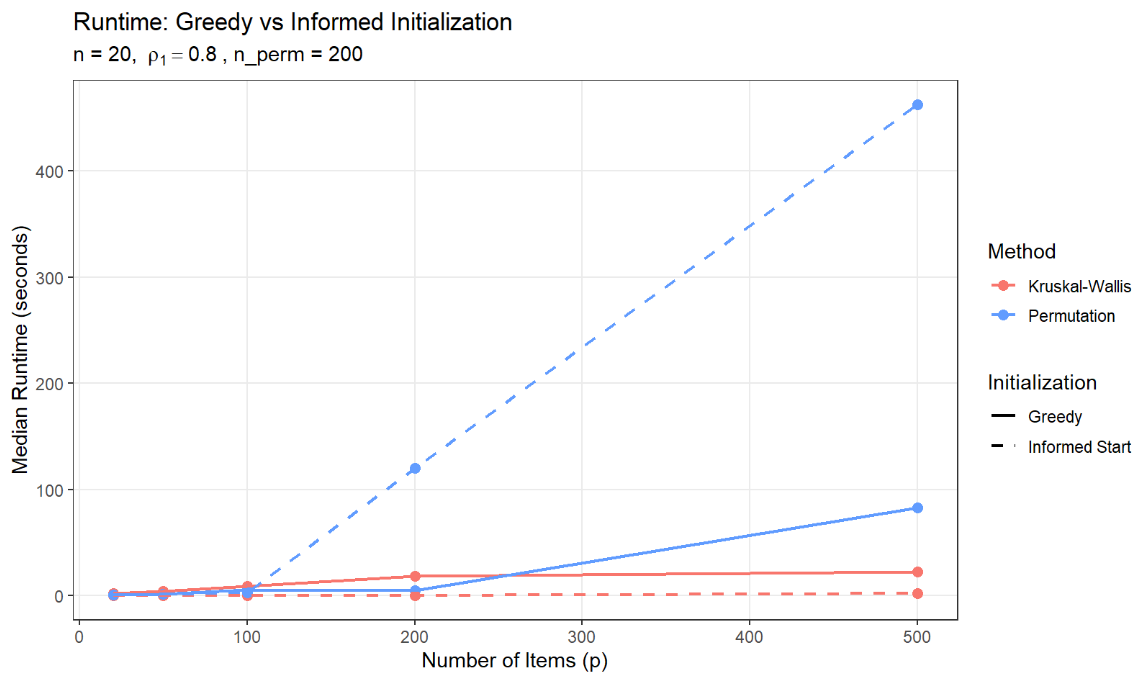

Figure 3.8

= 500 compared to roughly 83 seconds under greedy initialization. This is likely explained by the interplay between sensitivity and convergence behavior. Due to high sensitivity in the permutation method, starting from a strong, true DC set, likely causes constant updates of candidate set, requiring more iterations to converge. The greedy initialization, by contrast, produces shorter runtimes at large p not because it is more efficient per iteration, but because its reduced accuracy at high p, as seen in the simulation results, leads to earlier termination with a smaller returned set. In this sense, the greedy permutation runtime at large p reflects the method finding less rather than converging faster.

In terms of the memory cost of VSAT, at each permutation iteration, VSAT computes three $p \times p$ correlation matrices and discards them immediately before the next iteration begins, so peak memory in use at any moment is capped at six $p \times p$ matrices. In practice this means peak memory usage is $O(p^2)$ regardless of the number of permutations or samples, and is unlikely to cause issues on modern hardware for the item set size in typical applications. Problems would only be expected to arise for population sizes substantially exceeding $p =$

500.

3.4 Simulation Study Implecations

Overall, the simulation results paint a clear picture of the trade-offs between the two methods. The permutation method is the more sensitive choice, particularly when the differential signal is weak, when background correlation is present across all groups, or when sample sizes are small. However, this sensitivity comes at the cost of a higher false discovery rate under the null and substantially greater computational expense at large item population sizes (p). The Kruskal-Wallis method is faster, more conservative, and more precise at strong signal levels, but struggles to detect differential correlation when the signal is moderate or when there is baseline correlation across treatments. In practice, the results of this simulation study conveys the following: (1) When signal is expected to be strong and p is large, the Kruskal-Wallis method is the practical choice. (2) When signal may be weak or a background correlation structure is present, the permutation method is preferred provided reasonable p and available compute time permit.

Chapter 4

VINEYARD DATA

Recall the vineyard microbial data, introduced in Section 1 Watts et al. (n.d.), in which microorganism counts were recorded across soil samples collected over three years from an experimental plot in a vineyard. The study consisted of three experiments — Herbicide (2 treatment levels), Crop Cover (3 treatment levels), and Fertilizer (3 treatment levels) — each designed to assess how different vineyard management practices influence the soil microbial community. In each soil sample, counts for thousands of microbial species are recorded. Following Stoen, Tran, & Chan (2025), we aggregate these counts to the class taxonomic level and analyze beta diversity behavior across treatment conditions, as class-level patterns provide a more interpretable and stable signal than the sparse species-level counts.

Pairwise differential correlation analysis of this dataset has previously been conducted in Stoen et al. (2025), where DC sets were identified by applying a two-treatment clustering algorithm separately to each pair of treatment levels. While informative, pairwise comparisons cannot capture the overall structure of correlation differences across all three conditions simultaneously. An identified DC set between two specific levels may not have true differential correlation behavior across all three treatments.

We apply VSAT to each experiment in two ways. First, with no pre-specified initial set, allowing the greedy initialization procedure to determine the starting point. Second, using the DC sets identified from the pairwise comparisons in Stoen et al. (2025) as user-specified initial sets, which allows us to assess whether those previously identified sets hold up under the more stringent three-treatment framework and whether VSAT can refine or change them in a meaningful way.

4.1 Vineyard Data Results

Applying VSAT to the vineyard dataset with greedy initialization returned no differentially correlated sets under either method across both the Crop Cover and Fertilizer experiments. Given the simulation results indicating that VSAT reliably detects differential correlation for dimensions comparable to these vineyard data ($p = 72$, $n = 36$), these results suggest that differential correlation at the class taxonomic level is weak or absent in both experiments.

To further probe whether differential correlation exists in the vineyard data, we use the DC sets identified from pairwise analysis in Stoen et al. (2025) as pre-specified initial sets for VSAT. For each experiment, pairwise DC sets were available for all directed comparisons between treatment levels, yielding six sets of modules for both the Crop Cover and Fertilizer experiments. These were pooled into a single collection of starting sets and passed individually as A_0 to VSAT, allowing us to assess whether any pairwise-identified set holds up under the more stringent simultaneous three-treatment framework. The permutation method was used due to its greater sensitivity.

Passing all pairwise DC sets as pre-specified initial sets produced sparse results in both experiments, with the overwhelming majority returning empty under the three-treatment framework. Most of the starting modules converged to empty sets, 71 of the crop cover starting modules and 88 of the fertilizer starting modules, indicating that most pairwise-identified differential correlation does not persist when assessed across all three treatment levels.

In the Fertilizer experiment, three non-trivial returned sets of size two or greater were identified. The most reliable of these is caldilineae and sphingobacteriia, which was returned independently by three distinct starting modules, suggesting this pair represents the most robust differential correlation signal across fertilizer treatments. Two other sets — blastocatellia, gammaproteobacteria, and spartobacteria, and firmicutes and nitriliruptoria — were each reached by a single starting module.

Looking more deeply into the correlation between Caldilineae and Sphingobacteriia across the three fertilizer treatments, the pair shows a strong positive correlation under no fertilizer ($r = 0.684$), weak positive correlation under organic fertilizer ($r = 0.136$) and a weak negative correlation under synthetic fertilizer ($r = -0.092$). This pattern shows differential behavior between these two classes associated with fertilizer application.

The Crop Cover experiment produced more non-trivial returned sets, with 13 of the starting modules returning sets of size two or greater. The two most reliable are fibrobacteria and thermodesulfobacteria, and holophagae and mollicutes, each of which was returned by two independent starting modules. The remaining 11 sets were each reached by a single starting module.

Notably, across both the Crop Cover and Fertilizer experiments, the returned sets consistently differ from their starting modules, indicating that VSAT is not simply confirming the pairwise results but actively refining the set based on the three-treatment correlation structure. In several cases, classes that appeared in the pairwise starting set were dropped entirely from the returned set, while new classes not present in the original pairwise module were added, suggesting that some pairwise-identified differential correlation does not generalize to the three-treatment setting. This highlights the value of a simultaneous three-treatment framework, as pairwise analysis alone cannot distinguish differential correlation that will hold coherently across all conditions.

Chapter 5

CONCLUSION

This thesis introduced VSAT (Variable to Set Association Testing) for three or more treatments, a generalization of the Differential Correlation Mining framework of Bodwin et al. (2018) to the three-or-more treatment setting, motivated by the structure of the vineyard soil microbiome experiment. Two non-parametric testing approaches were developed — a Kruskal-Wallis method and a permutation-based method — after a parametric F-test approach was shown to be unreliable due to violated independence and normality assumptions in the correlation contribution framework. The simulation study established that both methods reliably recover true DC sets when within-set correlation is at least ρ_1 = 0.6 and sample sizes are adequate, with the permutation method preferred when signal is weak or background correlation is present, and the Kruskal-Wallis method preferred when signal is strong, p is large, or false discovery control is a priority.

Applied to the vineyard dataset, greedy initialization returned no DC sets in either experiment, suggesting that differential correlation at the class taxonomic level is weak or absent. Using pairwise DC sets from prior analysis as pre-specified initial sets revealed sparse but interpretable structure, with the most robust finding being a clear correlation gradient between Caldilineae and Sphingobacteriia across fertilizer treatments. The returned sets consistently differed from their starting modules, demonstrating that VSAT for three or more treatments finds meaningful patterns in three treatments simultaneously that differ from the pairwise results.

5.1 Future Work

Several directions for future work related to VSAT remain. The most natural next step, specifically for the vineyard data, would be to apply VSAT at a finer taxonomic level. Aggregating counts to the class level, while providing stability and a more manageable population size, may smooth over differential correlation patterns. Applying VSAT at the order or family level would increase the resolution of the analysis and may reveal new differential correlation behavior that is lost by aggregating.

The most impactful would be deriving the appropriate parametric null distribution for the test statistic using the contribution breakdown outlined in Section 2.2 and Section 2.3. While the Kruskal-Wallis method already scales reasonably well to large item sets, a parametric alternative to permutation testing would allow for exact p-values accessible and efficient for extremely large item sets. Related to deriving exact p-values for set determination, an important improvement noted in Section 3.2 would be developing formal Type I error control for VSAT, as both methods show elevated false discovery rates under null correlation.

Another possible improvement is the application of a variance stabilization technique, such as that used in (Gómez et al., 2025), to the correlation estimates prior to testing, which may reduce sensitivity to small correlation magnitudes and improve the performance of the permutation test application.

The initialization procedure also likely requires optimization. The current greedy approach is effective but carries relatively large cost at large p, and faster or more theoretically motivated starting points could improve runtime without sacrificing discovery.

Beyond these, application and formal definitions for more than three treatment groups is the natural next step, allowing for comparison under an arbitrary k number of treatment simultaneously.

5.2 Github Link

All the functions, testing and evaluation code used for this project are available at https://github.com/tylerstoen/VSAT.

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5.3 Computational Details

The results in this paper were obtained using `R` 4.6.0. `R` itself and all packages used are available from the Comprehensive `R` Archive Network (CRAN) at https://CRAN.R-project.org/.